

1. Find Mean, Median, Mode and Range

Data:

12, 15, 18, 15, 20, 22, 15, 25

Step 1: Arrange in ascending order

12, 15, 15, 15, 18, 20, 22, 25

Mean

$$\begin{aligned} \text{Mean} &= \frac{\sum X}{n} \\ &= \frac{12 + 15 + 18 + 15 + 20 + 22 + 15 + 25}{8} \\ &= \frac{142}{8} = \boxed{17.75} \end{aligned}$$

Median

$$\begin{aligned} \text{Median} &= \frac{4th + 5th}{2} \\ &= \frac{15 + 18}{2} = \boxed{16.5} \end{aligned}$$

Mode

15 occurs 3 times, which is the highest frequency.

Mode = 15

Range = Maximum - Minimum

Answer:

- Mean = 17.75
- Median = 16.5
- Mode = 15
- Range = 13

2. Construct a Frequency Distribution

Data:

2, 3, 4, 2, 5, 3, 4, 2, 6, 5, 3, 4

Count how many times each value occurs.

Value	Frequency
2	3
3	3
4	3
5	2
6	1
Total	12

Frequency Distribution : 2:3, 3:3, 4:3, 5:2, 6:1

3. Find Q1, Q2, Q3 and IQR

Data:

5, 8, 10, 12, 15, 18, 20, 22, 25, 30

There are 10 observations.

Q2 — Median

Q1

Lower half:

5, 8, 10, 12, 15

Middle value:

Q3

Upper half:

18, 20, 22, 25, 30

Middle value:

IQR

Answer:

- Q1 = 10

- $Q2 = 16.5$
- $Q3 = 22$
- $IQR = 12$

4. Detect Outliers Using IQR Method

Data:

10, 12, 13, 15, 16, 17, 18, 20, 45

Q1

Lower half:

10, 12, 13, 15

Q3

Upper half:

17, 18, 20, 45

IQR

Lower Bound

Upper Bound

Any value below 2.75 or above 28.75 is an outlier.

Since: $45 > 28.75$

45 is an outlier

5. Determine the Skewness

Data:

5, 6, 7, 7, 8, 8, 9, 20

The value 20 is much larger than the other observations.

$$\begin{aligned} \text{Mean} &= \frac{5 + 6 + 7 + 7 + 8 + 8 + 9 + 20}{8} \\ &= \frac{70}{8} = 8.75 \end{aligned}$$

Median

$$\text{Median} = \frac{7 + 8}{2} = 7.5$$

Middle two values are 7 and 8.

The mean is greater than the median: $8.75 > 7.5$

Therefore, the distribution has a long right tail.

Positively skewed / Right-skewed

Reason: The extreme value 20 pulls the mean toward the right.

6. Uniform Distribution

A uniformly distributed variable ranges from 10 to 50.

Find: $P(20 < X < 35)$

For a uniform distribution: $P(a < X < b) = \frac{\text{Maximum} - \text{Minimum}}{b - a}$

Therefore:

$$\begin{aligned} P(20 < X < 35) &= \frac{35 - 20}{50 - 10} \\ &= \frac{15}{40} \end{aligned}$$

$$P(20 < X < 35) = 0.375$$

$$37.5\%$$

7. Find Z-Score

Given:

- Mean $\mu = 100$
- Standard deviation $\sigma = 15$
- $X = 130$

$$Z = \frac{X - \mu}{\sigma}$$
$$Z = \frac{130 - 100}{15}$$
$$Z = \frac{30}{15}$$
$$\boxed{Z = 2}$$

Formula:

So, 130 is 2 standard deviations above the mean.

8. Find $P(X < 130)$

From Question 7: $Z = 2$

Using the standard normal distribution table:

$$P(Z < 2) = 0.9772$$

$$\boxed{P(X < 130) = 0.9772}$$

$$\boxed{97.72\%}$$

9. Compare Mean and Median Before and After Outlier Treatment

Consider the dataset:

10, 12, 13, 14, 15, 16, 100

Here, 100 is an extreme outlier.

Before Outlier Treatment

Mean

Median

The middle value is:

So:

Measure	Before Treatment
Mean	25.71
Median	14

The mean is strongly affected by the outlier 100.

After Removing the Outlier

New dataset:

10, 12, 13, 14, 15, 16

Mean

Median

Measure	Before	After
Mean	25.71	13.33
Median	14	13.5

Conclusion

The mean changes significantly because it is sensitive to outliers, while the median changes only slightly.

10. Complete EDA on a Student-Mark Dataset

Let's use this small student-mark dataset:

45, 52, 56, 60, 62, 65, 68, 70, 72, 95

A. Dataset

Student	Marks
S1	45
S2	52
S3	56
S4	60
S5	62

S6	65
S7	68
S8	70
S9	72
S10	95

Number of students: n=10

B. Central Tendency

Mean

$$\text{Mean} = \frac{45 + 52 + 56 + 60 + 62 + 65 + 68 + 70 + 72 + 95}{10}$$

$$\text{Mean} = 64.5$$

Median

$$\text{Median} = \frac{62 + 65}{2}$$

$$\text{Median} = 63.5$$

Middle two values:

62 and 65

Mode

No mode

Every mark occurs only once.

Central Tendency Summary

Measure	Value
Mean	64.5
Median	63.5

Mode	No mode
------	---------

C. Dispersion

$$\text{Range} = 95 - 45$$

$$\text{Range} = 50$$

Range :

Q1

Lower half:

45, 52, 56, 60, 62 Q1 =56

Q3

Upper half:

65, 68, 70, 72, 95 Q3 =70

IQR

IQR =70-56

IQR = 14

Standard Deviation: SD = 13.60

Sample standard deviation is approximately:

Variance: variance = 184.94

Dispersion Summary

Measure	Value
Range	50
Q1	56
Q2	63.5
Q3	70
IQR	14
Standard Deviation	13.60
Variance	184.94

D. Outlier Detection

Using the IQR method: $Q1 = 56$, $Q3 = 70$

$IQR = 14$

Lower Bound: $56 - 1.5(14) = 56 - 21 = 35$

Upper Bound: $70 + 1.5(14) = 70 + 21 = 91$

Therefore, acceptable values lie between: $35 \leq X \leq 91$

The mark 95 is greater than 91.

Therefore: 95 is an outlier

E. Distribution and Skewness

Mean: 64.5

Median: 63.5

Since: Mean > Median

the distribution is positively/right skewed.

The calculated skewness is approximately: 0.88

Therefore, the student marks show moderate positive skewness, mainly because of the high mark 95.

Final EDA Report

Category	Result
Number of students	10
Mean	64.5
Median	63.5
Mode	No mode
Minimum	45
Maximum	95
Range	50
Q1	56

Q2	63.5
Q3	70
IQR	14
Standard Deviation	13.60
Variance	184.94
Outlier	95
Skewness	+0.88
Distribution	Right/positively skewed

Overall Conclusion

The students' average mark is 64.5, while the median is 63.5. The data has a relatively large range of 50 marks. The value 95 is identified as an outlier using the IQR method. Because of this high value, the distribution is positively skewed, and the mean is slightly higher than the median.